



MATHEMATICS · YEAR 7 · HOMEWORK
BOOKLET

Chapter 8: Area and volume

Mixed revision of all 13 lessons · 32 questions

NSW syllabus outcome

Stage 4 Measurement and space: MA4-LEN-C-01 (length and circumference), MA4-ARE-C-01 (area) and MA4-VOL-C-01 (volume and capacity).

NAME

CLASS



Mixed revision homework. These questions mix all the lessons in Chapter 8, from **Easy** to **Challenging**. Each one is tagged with its lesson, so you can look back at that lesson's summary if you get stuck. Show your working in the boxes.

YOU DO

On your own: mixed revision.

QUESTIONS 1–10 OF 32

1 **Easy**

Convert.

2.5 L to mL Lesson 8.01

2 **Easy**

Find the circumference (1 d.p.).

$d = 8$ m Lesson 8.03

3 **Easy**

Find the area.

rectangle 9 m by 8 m Lesson 8.05

4 **Easy**

Find the area of the parallelogram.

$b = 13$ m, $h = 5$ m Lesson 8.07

5 **Easy**

Name the shape of the cross-section.

octagonal prism Lesson 8.09

6 **Easy**

Find the volume.

7 m $\times$ 4 m $\times$ 3 m Lesson 8.11

7 **Easy**

Convert.

4 m³ to L Lesson 8.13

8 **Easy**

Find the perimeter.

rectangle 9 m by 6 m Lesson 8.02

9 **Easy**

Convert.

6 m² to cm² Lesson 8.04

10 **Easy**

Find the area of the triangle.

$b = 14$ cm, $h = 6$ cm Lesson 8.06

YOU DO

On your own: mixed revision.

QUESTIONS 11–20 OF 32

11

Easy

Find the area.

a 12×5 rectangle with a 3×3 hole (cm) Lesson 8.08

12

Easy

Convert.

7 cm^3 to mm^3 Lesson 8.10

13

Easy

Find the volume.

$A = 22 \text{ m}^2$, $h = 5 \text{ m}$ Lesson 8.12

14

Medium

Convert 2750 g to kg. Lesson 8.01

15

Medium

Find the circumference of a circle with diameter 9 m (1 d.p.). Lesson 8.03

16

Medium

Find the area of a square with side 2.5 m. Lesson 8.05

17

Medium

Find the area: base 2.5 m, height 1.6 m. Lesson 8.07

18

Medium

Name the cross-section of a hexagonal prism. Lesson 8.09

19

Medium

Find the volume of a cube with edge 6 m. Lesson 8.11

20

Medium

Find the capacity in litres of a box $25 \text{ cm} \times 20 \text{ cm} \times 10 \text{ cm}$. Lesson 8.13

YOU DO On your own: mixed revision.

QUESTIONS 21–30 OF 32

- 21 **Medium** Find the perimeter of an equilateral triangle with sides 9.5 m. Lesson 8.02

- 23 **Medium** Find the area: base 5 cm, height 3.2 cm. Lesson 8.06

- 25 **Medium** Convert 2 500 000 cm³ to m³. Lesson 8.10

- 27 **Challenging** Convert 3.5 h to minutes. Lesson 8.01

- 29 **Challenging** A rectangle has area 60 m² and width 5 m. Find its length. Lesson 8.05

- 22 **Medium** Convert 30 000 cm² to m². Lesson 8.04

- 24 **Medium** Find the area of an L-shape made from a 6 m × 3 m and a 2 m × 2 m rectangle. Lesson 8.08

- 26 **Medium** Triangle base 5 m, height 4 m, prism length 9 m. Find the volume. Lesson 8.12

- 28 **Challenging** Find the circumference of a circle with radius 3.5 cm (1 d.p.). Lesson 8.03

- 30 **Challenging** A parallelogram has area 50 m² and height 4 m. Find its base. Lesson 8.07

YOU DO

On your own: mixed revision.

QUESTIONS 31–32 OF 32

- 31** **Challenging** Sketch the top view of a triangular prism lying on its rectangular face. Lesson 8.09

- 32** **Challenging** A box has volume 72 cm^3 and base area 12 cm^2 . Find its height. Lesson 8.11

YOU DO

On your own: extra working space. Write the question number next to your working.